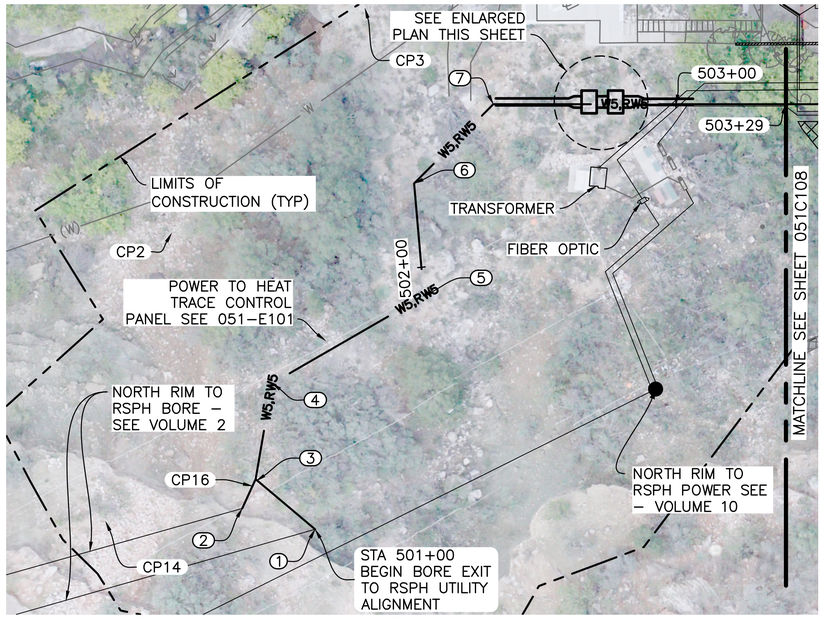
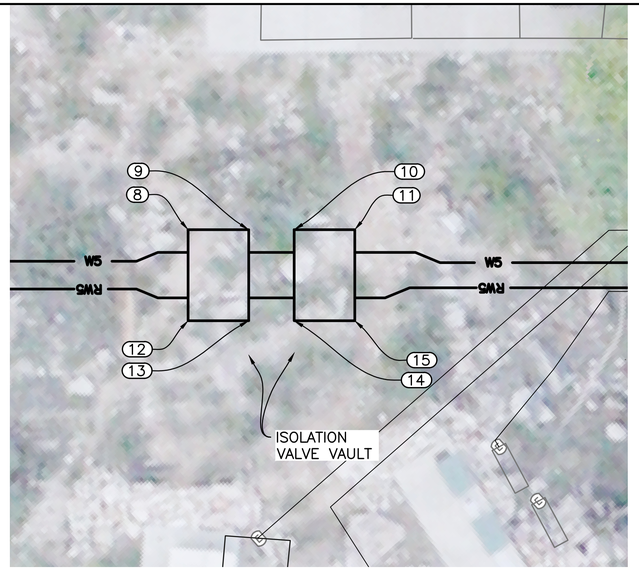
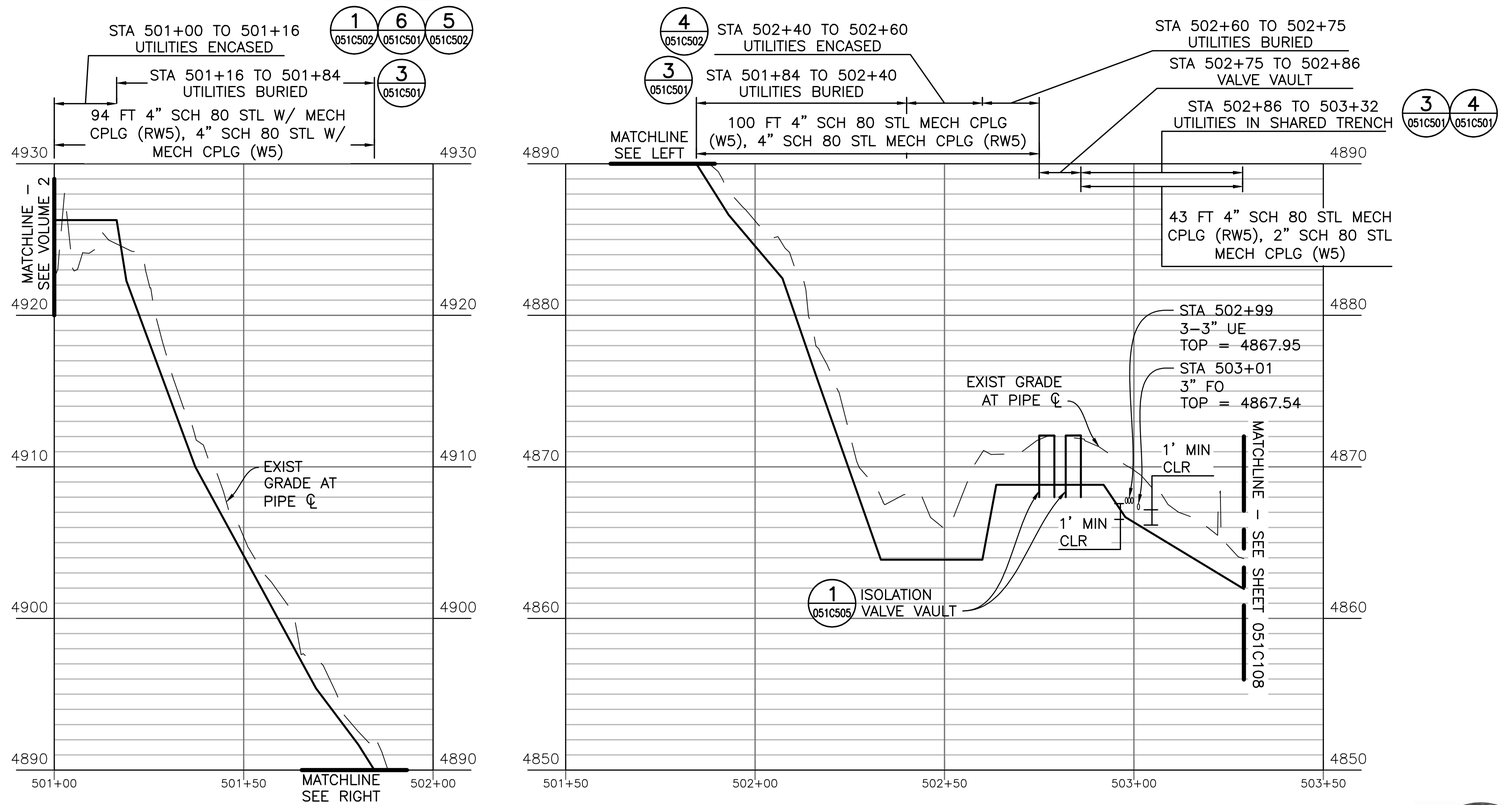


PLAN



ENLARGED PLAN

SCALE: 1" = 5'



PROFILE A – BORE TO RSPH PROFILE

GENERAL NOTES:

A. SURVEY CONTROL AND CONTROL POINT TABLE LOCATED IN VOLUME 0, SEE SHEET 002C105.

B. AVAILABLE CONSTRUCTION WIDTH IS INTENDED TO DEFINE THE LIMITS OF CONSTRUCTION CONSIDERING SITE TOPOGRAPHY, INCLUDING TRAIL WIDTH, AND MAXIMUM ALLOWABLE DISTURBANCE.

B. LENGTHS OF PIPE SHOWN IN PROFILE REFLECTS CALCULATED SLOPE LENGTHS.

D. NORTHINGS AND EASTINGS SHOWN FOR SHARED UTILITY ALIGNMENTS REFLECT TRENCH CENTERLINE.

E. MINOR DEVIATIONS IN THE NEW PIPELINE ALIGNMENT AND PROFILE ARE ANTICIPATED TO FACILITATE CONSTRUCTION. SEE SPECIFICATIONS FOR ALIGNMENT AND PROFILE REQUIREMENTS, MITERED JOINTS AND TRIMMED FITTINGS.

F. REFER TO SPECIFICATION 32 32 16 – PRECAST MODULAR BLOCK RETAINING WALL FOR MODULAR DESIGN AND CONSTRUCTION.

G. GROUND SURFACE BASED ON PRE–FLOOD SURVEY. CONTRACTOR TO FIELD VERIFY EXISTING SURFACE.

KEYNOTES ⟨X⟩

1. BORE EXIT LOCATIONS SHOWN ARE APPROXIMATE. CONTRACTOR TO MAKE FIELD ADJUSTMENTS BASED ON CONSTRUCTED BORE EXIT LOCATIONS.

WATERLINE FITTING TABLE – BORE EXIT TO RSPH

POINT NO.	NORTHING	EASTING	DESCRIPTION
1	13142710.715	1335205.377	CONNECT TO BORE LINE ⟨1⟩
2	13142727.689	1335194.799	CONNECT TO BORE LINE ⟨1⟩
3	13142730.628	1335202.426	1–4" 60° BEND, 1–4" 15° BEND
4	13142746.898	1335222.809	1–4" 50° BEND, 1–4" 50° BEND
5	13142737.262	1335267.767	1–4" 65° BEND, 1–4" 65° BEND
6	13142760.962	1335285.953	1–4" 50° BEND, 1–4" 50° BEND
7	13142762.430	1335315.494	1–4" 45° BEND, 1–4" 45° BEND
8	13142749.489	1335335.218	VAULT CORNER
9	13142746.807	1335338.186	VAULT CORNER
10	13142744.795	1335340.411	VAULT CORNER
11	13142742.113	1335343.379	VAULT CORNER
12	13142745.037	1335331.195	VAULT CORNER
13	13142742.355	1335334.163	VAULT CORNER
14	13142740.344	1335336.388	VAULT CORNER
15	13142737.662	1335339.356	VAULT CORNER

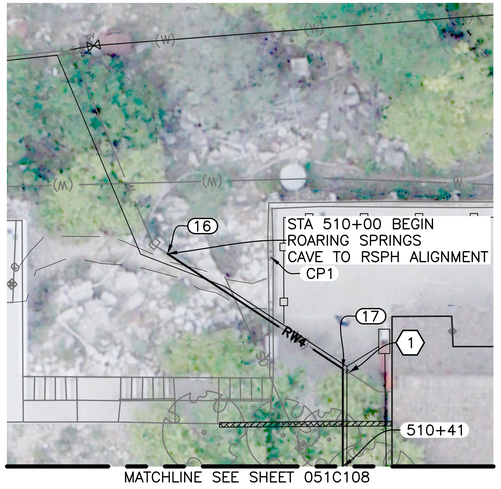
20 0 20 40 HORIZ.
5 0 5 10 VERT.
SCALE OF FEET
H: 1' = 20' V: 1" = 5'

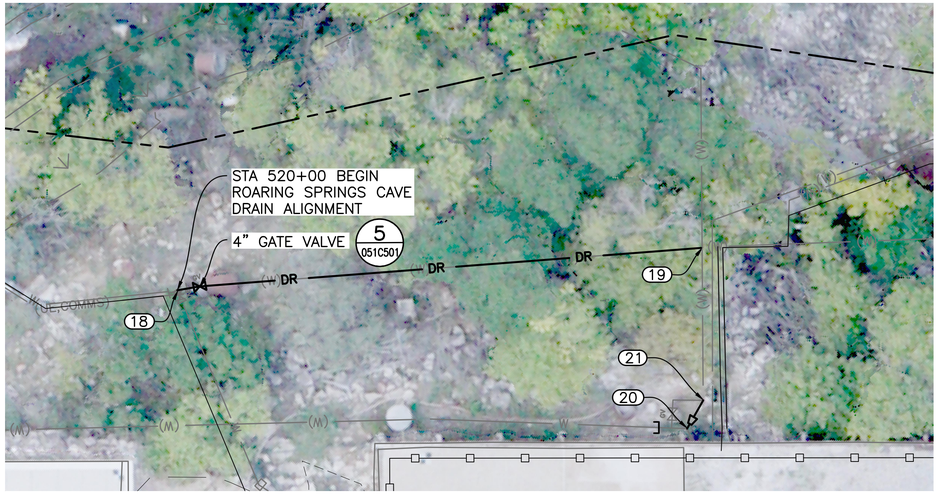
Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 10-9-2026 Date Signed ARIZONA, U.S.A.

Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: S. RACE DRAWN: C. WITTENMEIER TECH. REVIEW: GOUGH/GLOVER DATE: JUNE 2026	SUB SHEET NO. 051 C104	TITLE OF SHEET SITEWORK PLAN AND PROFILE A NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 14 OF 113

10/12/2026 Green, Jake C:\Users\jagreen\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\CIVIL\051-C104.dwg





GENERAL NOTES:

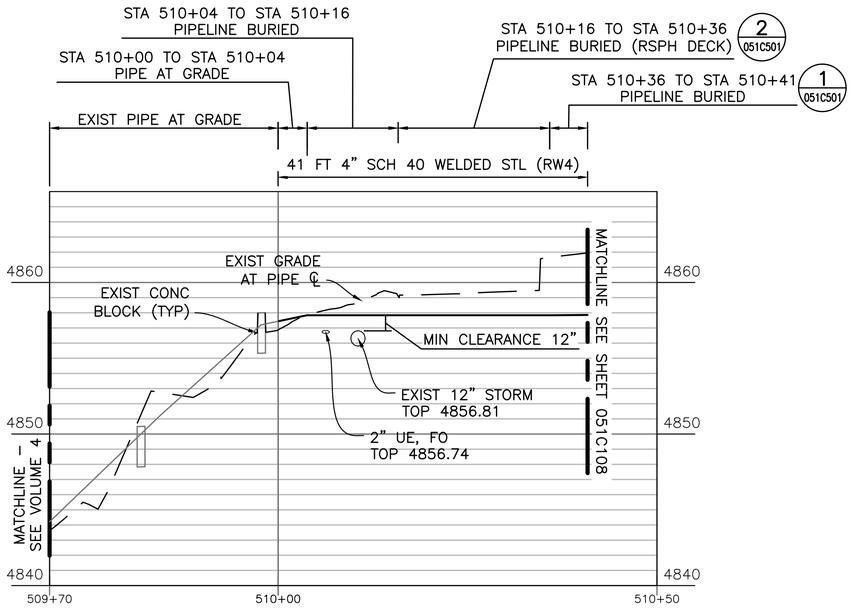
A. SURVEY CONTROL AND CONTROL POINT TABLE LOCATED IN VOLUME 0. SEE SHEET 002C105.

B. AVAILABLE CONSTRUCTION WIDTH IS INTENDED TO DEFINE THE LIMITS OF CONSTRUCTION CONSIDERING SITE TOPOGRAPHY, INCLUDING TRAIL WIDTH, AND MAXIMUM ALLOWABLE DISTURBANCE.

B. LENGTHS OF PIPE SHOWN IN PROFILE REFLECTS CALCULATED SLOPE LENGTHS.

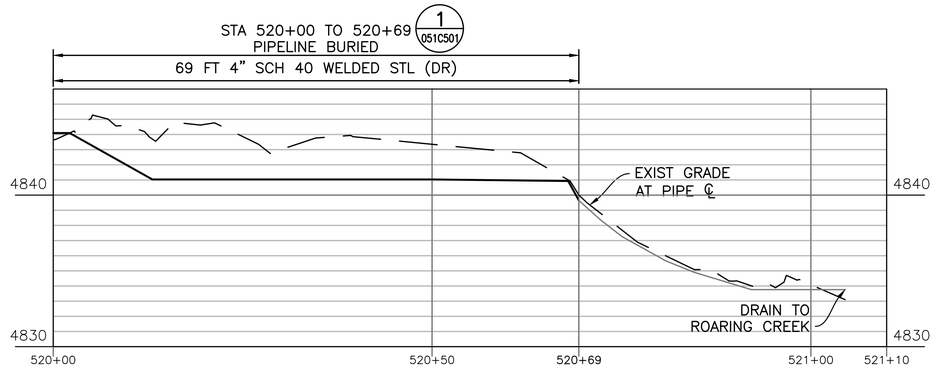
KEYNOTES ⟨X⟩

1. EXISTING PIPE TO BE CAPPED.



PROFILE B – RS CAVE TO RSPH PROFILE

WATERLINE FITTING TABLE – RS CAVE TO RSPH			
POINT NO.	NORTHING	EASTING	DESCRIPTION
16	13142752.67	1335383.64	CONNECT TO EXISTING 8" STL PIPE W/ 8"X4" REDUCER
17	13142725.89	1335390.62	FIELD MEASURE AND FABRICATE 4" BEND TO TIE–IN TO ALIGNMENT



PROFILE C – RS CAVE DRAIN PROFILE

WATERLINE FITTING TABLE – RS CAVE DRAIN			
POINT NO.	NORTHING	EASTING	DESCRIPTION
18	13142781.10	1335392.31	CONNECT TO EXISTING TEE W/ 8"X4" REDUCER, 4" GV, INSULATING KIT
19	13142738.87	1335447.48	4"X4" TEE, CONNECT TO EXIST WITH 1" ORIFICE PLATE
20	13142722.50	1335430.02	FIELD MEASURE AND FABRICATE 6" BEND AND 6"X4" REDUCER TO TIE–IN TO EXIST 6" STL WATERLINE
21	13142723.85	1335434.10	FIELD MEASURE AND FABRICATE 4" BEND AND TIE–IN TO EXIST 4" STL DRAIN

10 0 10 20 HORIZ.
5 0 5 10 VERT.
SCALE OF FEET
H: 1" = 10' V: 1" = 5'

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.

Nicholas Humphrey

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	S. RACE	051 C105	SITEWORK PLAN AND PROFILE B+C	113 182752
	DRAFTED: C. WITTENMEIER			PMIS/PKG NO. 318719
	TECH. REVIEW: GOUGH/GLOVER		NORTH RIM - GRAND CANYON NATIONAL PARK	SHEET 15 OF 113
	DATE: MAY 2026			

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